

CertaRig Phase 3

Wave 1 Wiring and Circuit Guide

Raspberry Pi 3 Model A+ dry hardware-in-the-loop bench
Prepared for controlled assembly, commissioning evidence, and the 20 September Phase 3 submission

CONTROL STATUS ANALOGUE INPUT PATH VERIFIED

Pi boot, SSH, I2C, ADS1115 address 0x48, and both 0-3.3 V potentiometer acquisition paths have measured evidence. E-stop contacts, transistor pinout, relay logic, fused output supply, and indicator polarity remain gated until measured and recorded.

Field	Value
Project	CertaRig
Scope	Wave 1 dry low-voltage bench only
Controller	Raspberry Pi 3 Model A+
Analogue interface	ADS1115 at 3.3 V, I2C address 0x48
Safety concept	NC E-stop sense loop plus independent NC relay-power inhibit
Output	Low-voltage green/yellow indicators through relay channel 1
Excluded	Mains, pump, solenoid valve, pressurised plumbing, water loop

1. Purpose and non-negotiable boundaries

This document is the single wiring reference for the CertaRig Wave 1 dry bench. It translates the received inventory and the current repository configuration into a labelled circuit, a staged assembly method, and evidence gates. It does not claim that the hardware is already electrically verified.

SAFE SCOPE

Use only SELV low-voltage sources: the official Raspberry Pi supply and a separate regulated 5 V supply for the relay and indicators. Do not connect mains voltage, a pump, a solenoid valve, or any water-bearing equipment during Wave 1.

STOP CONDITIONS

Stop immediately for heat, smell, smoke, unstable Pi power, an unexpected voltage, unclear terminal labels, a non-latching E-stop, an indicator that does not match the state, or any software state that disagrees with the physical state.

What Wave 1 can prove

- Real Pi deployment, I2C acquisition, analogue scaling, deterministic safety logic, dashboard/API behaviour, event logs, and repeatable test evidence.
- Healthy, warning, threshold-trip, E-stop, broken-wire, restart, and output-indication behaviours on a low-voltage dry bench.
- A controlled substitution boundary: potentiometers emulate sensor voltages and indicators emulate an actuator command.

What Wave 1 cannot prove

- Pressure or flow accuracy, leaks, priming, cavitation, surge, pump inrush, hydraulic dynamics, EMI in a field installation, or long-duration wet operation.
- Formal TRL certification. The defensible wording is evidence consistent with laboratory breadboard readiness.

2. System architecture and power domains

Wave 1 system architecture

Two power domains, one common reference, deterministic software and a hardwired inhibit

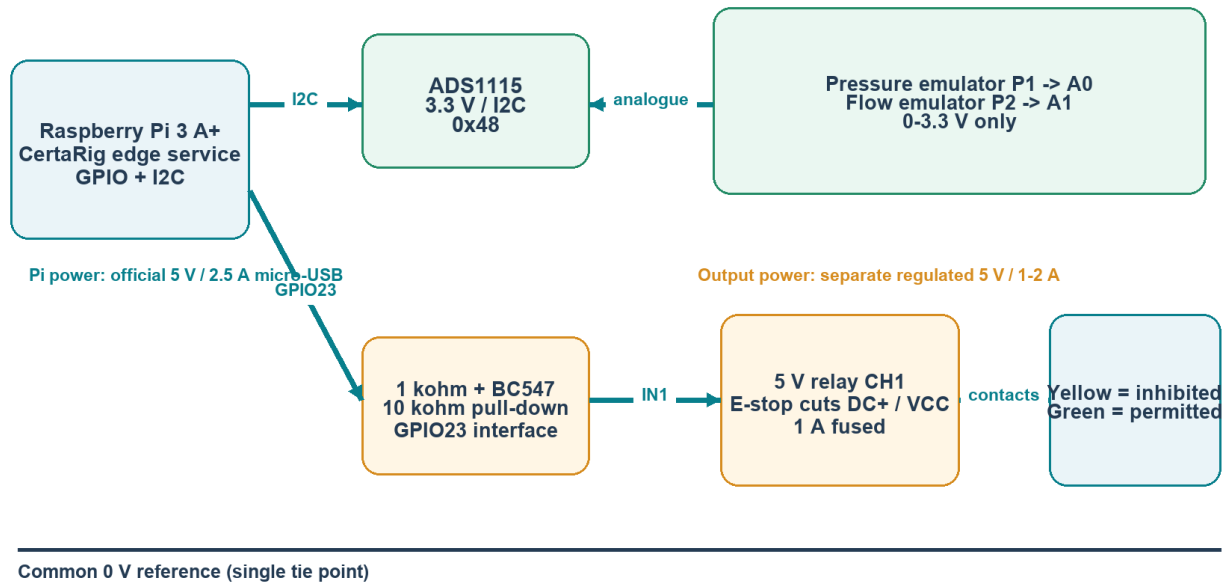


Figure 1. Wave 1 architecture. The Pi and output circuit use separate supplies; only the reference ground is shared for the transistor interface.

Domain	Source	Loads	Rule
Pi / logic	Official 5 V, 2.5 A micro-USB supply into PWR IN	Pi, ADS1115, two potentiometers	ADS uses physical pin 1; pots use physical pin 17 via the verified splitter
Output	Separate regulated 5 V, 1-2 A supply	Relay module and two 3-9 V indicators	Pass +5 V through the 1 A fuse and E-stop power loop
Common reference	One controlled ground tie	Pi GND, transistor emitter, relay/external GND	Never join the external +5 V rail to Pi 5 V pins

Why two supplies

The relay board can draw far more current and create switching noise. A separate supply keeps that load away from the Pi rail, while a single ground tie gives the BC547 interface a defined reference.

3. Raspberry Pi GPIO naming

Every connection in this guide uses both the physical header pin and the BCM GPIO number. Physical pin 16 and BCM GPIO23 are the same conductor; they are not two different pins. Never wire by BCM number alone while the Pi is unpowered.

Raspberry Pi 3 Model A+ 40-pin header

Abstract header view. Use physical pin number first; BCM GPIO number second.

ODD			EVEN		
1 3V3	USED: ADS VDD	●	2 5V		DO NOT USE for relay supply
3 GPIO2 / SDA1	USED: ADS SDA	●	4 5V		DO NOT USE for relay supply
5 GPIO3 / SCL1	USED: ADS SCL	●	6 GND		USED: ADS ground
7 GPIO4	Spare	○	8 GPIO14 / TXD		Reserved diagnostics
9 GND	USED: P2 analogue-pot ground	○	10 GPIO15 / RXD		Reserved diagnostics
11 GPIO17	Spare	○	12 GPIO18 / PWM0		Spare
13 GPIO27	Spare	○	14 GND		USED: P1 analogue-pot ground
15 GPIO22	Spare	○	16 GPIO23		USED: relay command
17 3V3	USED: analogue-pot 3.3 V feed	○	18 GPIO24		USED: E-stop sense
19 GPIO10 / MOSI	Spare	○	20 GND		USED: E-stop/common ground
21 GPIO9 / MISO	Spare	○	22 GPIO25		RESERVED: safe relay feedback
23 GPIO11 / SCLK	Spare	○	24 GPIO8 / CE0		Spare
25 GND	Spare ground	○	26 GPIO7 / CE1		Spare
27 GPIO0 / ID_SD	Do not use	○	28 GPIO1 / ID_SC		Do not use
29 GPIO5	Spare	○	30 GND		Spare ground
31 GPIO6	Spare	○	32 GPIO12 / PWM0		Spare
33 GPIO13 / PWM1	Spare	○	34 GND		Spare ground
35 GPIO19 / PCM_FS	Spare	○	36 GPIO16		Spare
37 GPIO26	Spare	○	38 GPIO20 / PCM_DIN		Spare
39 GND	Spare ground	○	40 GPIO21 / PCM_DOUT		Spare

Never apply 5 V to a GPIO or ADS1115 input. Pins 2 and 4 are not the external relay-supply connection.

Figure 2. Abstract 40-pin header map. Confirm orientation from the square pin-1 pad and later with the Raspberry Pi pinout command.

Used pins

Pin 1 = 3V3; pin 3 = GPIO2/SDA1; pin 5 = GPIO3/SCL1; pin 6 = GND; pin 16 = GPIO23 relay command; pin 18 = GPIO24 E-stop input; pin 20 = GND. Pin 22/GPIO25 is reserved but left unconnected for the first build.

4. Complete 40-pin allocation table

Phys	Odd pin	Type	CertaRig use	Phys	Even pin	Type	CertaRig use
1	3V3	POWER	USED: ADS VDD	2	5V	POWER	DO NOT USE for relay supply
3	GPIO2 / SDA1	I2C	USED: ADS SDA	4	5V	POWER	DO NOT USE for relay supply
5	GPIO3 / SCL1	I2C	USED: ADS SCL	6	GND	GROUND	USED: ADS ground
7	GPIO4	GPIO	Spare	8	GPIO14 / TXD	UART	Reserved diagnostics
9	GND	GROUND	USED: P2 analogue-pot ground	10	GPIO15 / RXD	UART	Reserved diagnostics
11	GPIO17	GPIO	Spare	12	GPIO18 / PWM0	GPIO	Spare
13	GPIO27	GPIO	Spare	14	GND	GROUND	USED: P1 analogue-pot ground
15	GPIO22	GPIO	Spare	16	GPIO23	OUTPUT	USED: relay command
17	3V3	POWER	USED: analogue-pot 3.3 V feed	18	GPIO24	INPUT	USED: E-stop sense
19	GPIO10 / MOSI	SPI	Spare	20	GND	GROUND	USED: E-stop/common ground
21	GPIO9 / MISO	SPI	Spare	22	GPIO25	INPUT	RESERVED: safe relay feedback
23	GPIO11 / SCLK	SPI	Spare	24	GPIO8 / CE0	SPI	Spare
25	GND	GROUND	Spare ground	26	GPIO7 / CE1	SPI	Spare
27	GPIO0 / ID_SD	HAT ID	Do not use	28	GPIO1 / ID_SC	HAT ID	Do not use
29	GPIO5	GPIO	Spare	30	GND	GROUND	Spare ground
31	GPIO6	GPIO	Spare	32	GPIO12 / PWM0	GPIO	Spare
33	GPIO13 / PWM1	GPIO	Spare	34	GND	GROUND	Spare ground
35	GPIO19 / PCM_FS	GPIO	Spare	36	GPIO16	GPIO	Spare
37	GPIO26	GPIO	Spare	38	GPIO20 / PCM_DIN	GPIO	Spare
39	GND	GROUND	Spare ground	40	GPIO21 / PCM_DOUT	GPIO	Spare

GPIO electrical rule: Raspberry Pi GPIO operates at 3.3 V. GPIO pins and ADS inputs must never receive 5 V. Power pins 2 and 4 are 5 V rails, not general outputs and not the external relay supply connection.

5. Circuit A: ADS1115 and sensor-emulation inputs

Circuit A Pi, ADS1115 and two sensor emulators

All analogue values stay inside the ADS1115 3.3 V supply rails

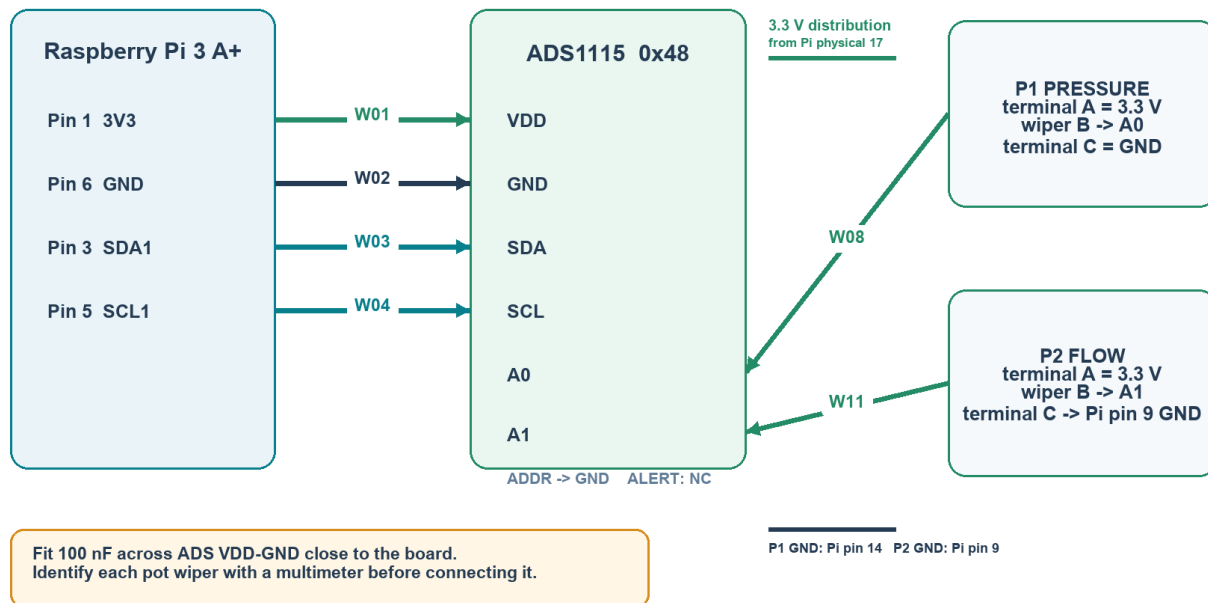


Figure 3. Pi-to-ADS1115 I2C wiring and two 10 kohm potentiometer sensor emulators.

Connection intent

- Power the ADS1115 from 3.3 V so its input range is naturally bounded to the Pi logic domain.
- Tie ADDR to GND for decimal address 72, hexadecimal 0x48. Leave ALERT/RDY unconnected during initial commissioning.
- Use A0 for pressure and A1 for flow. Reserve A2/A3 until the two-channel baseline is stable.
- Place one 100 nF ceramic capacitor directly across ADS VDD and GND.
- Characterise each potentiometer separately. In the measured Certarig convention, A is the high track end, B is the wiper, and C is the low track end; manufacturer markings and measurements override this illustration.

WAVE 1 CONFIGURATION

Use the dedicated Wave 1 configuration with measured endpoints near 0-3.3 V. The original 0.5-4.5 V example is retained only as a future sensor-interface reference and must not be used for the present potentiometer bench.

5.1 ADS1115 physical placement on the MB102 breadboard

ADS1115 physical placement on an MB102 830 point breadboard

Use the central five-hole terminal strips; do not use the long side power rails at this stage

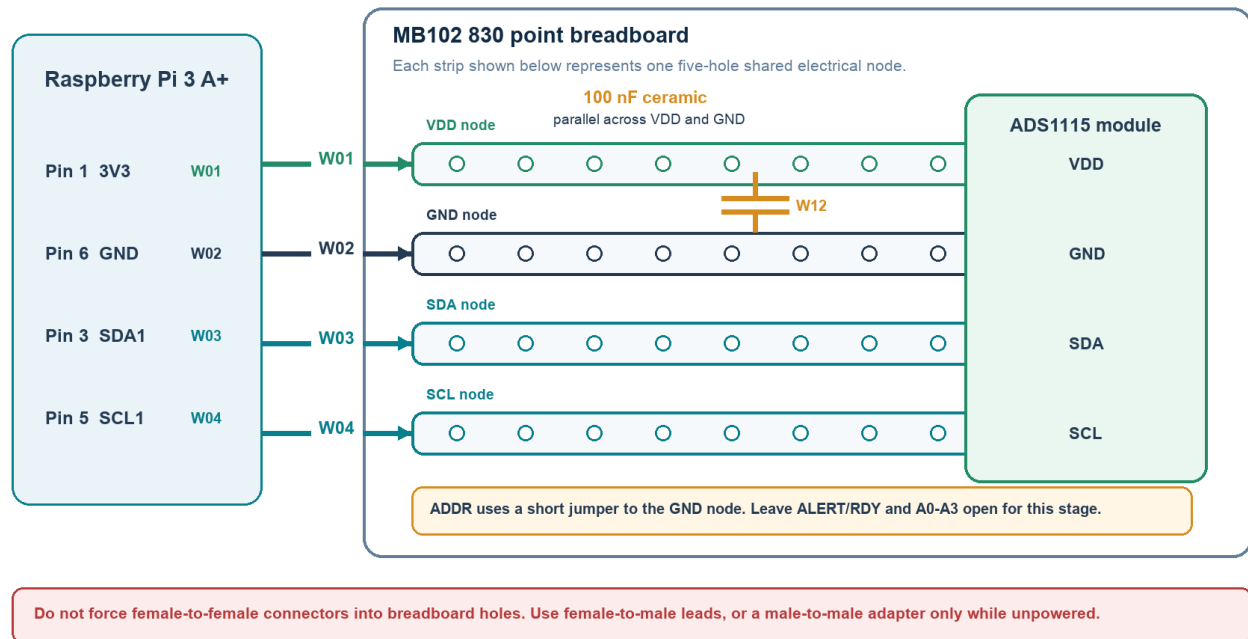


Figure 3A. Physical node layout for the ADS-only commissioning stage on an MB102 830 point breadboard.

How to assemble this stage

- Use the MB102 central terminal area, not the long red and blue side rails. On a typical MB102, holes A-E in one numbered row form one five-hole node, and holes F-J form a separate node across the centre channel. Verify this pattern with the multimeter because board variants differ and the long power rails may be split.
- Insert the ADS1115 male header along one terminal-strip bank so every labelled module pin occupies a different numbered five-hole row. Never place two ADS pins in the same connected row. Follow the ADS board silkscreen because breakout-board pin order varies.
- Use female-to-male jumper leads: the female end fits the Raspberry Pi GPIO header and the male end enters the breadboard strip shared with the matching ADS1115 pin.
- Place one leg of the 100 nF ceramic capacitor in the same strip as ADS VDD and the other leg in the same strip as ADS GND. The capacitor is parallel with the supply, not inserted in series with either wire, and it has no polarity.
- Connect ADS ADDR to the same ground strip with a short male-to-male jumper to select address 0x48.
- Leave ALERT/RDY and analogue inputs A0-A3 open. Do not connect the potentiometers until ADS-only detection has passed.

Direct female-to-female alternative. The four Pi-to-ADS connections can be made directly only if the external capacitor is omitted or attached through a purpose-built splitter. Do not twist the capacitor around exposed header pins. For this controlled build, use the breadboard arrangement above.

6. Circuit B: E-stop, relay interface, fuse and indicators

Circuit B fail-safe command, relay power and indicators

Dry low-voltage test only. No pump, mains voltage or water loop in Wave 1.

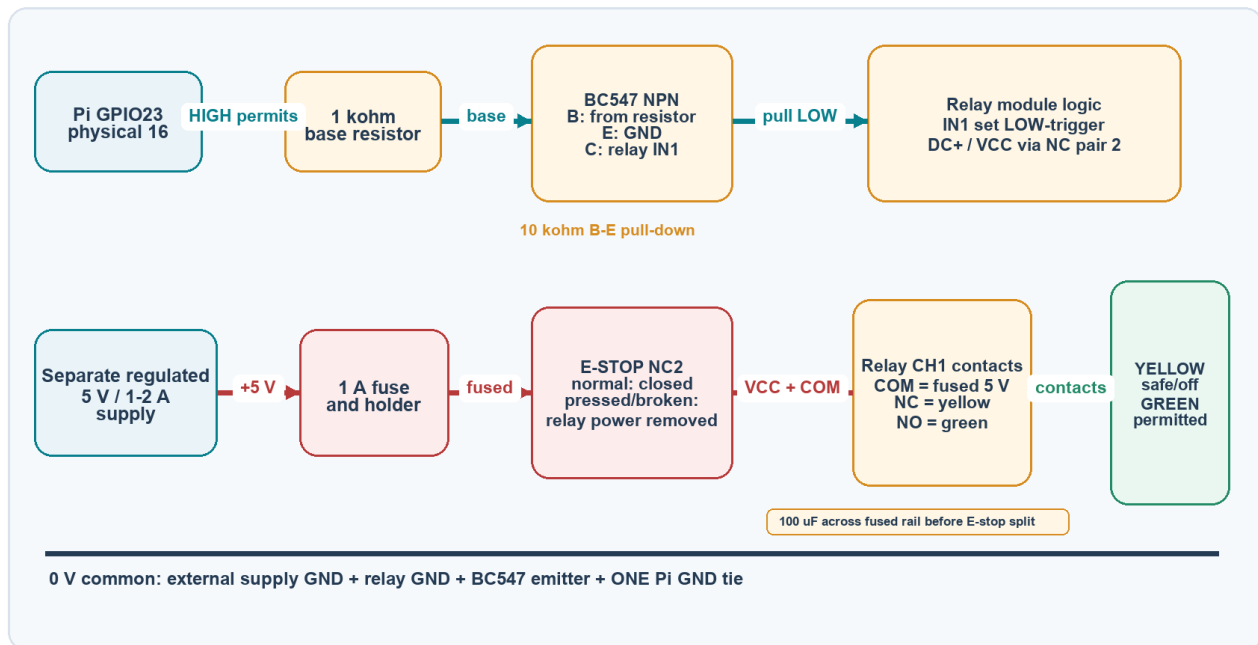


Figure 4. Intended fail-safe low-voltage output path. Exact module terminal order remains a measurement gate.

Dual-contact E-stop allocation

Contact set	Circuit	Normal condition	Pressed or broken wire	Purpose
Measured NC pair 1	GPIO24 to GND sense loop	Closed; input LOW	Open; input HIGH	Software detects E-stop and wiring break
Measured NC pair 2	Fused +5 V to relay DC+ or VCC	Closed; relay can energize	Open; relay power removed	Independent hardware inhibit

FAIL-SAFE POLARITY CONTRACT

The Wave 1 adapter uses a pull-up input with explicit active-high fault interpretation. The measured NC loop therefore has this electrical truth: normal/closed is LOW and healthy; pressed, open, or broken-wire is HIGH and active. Automated tests must pass before connecting the E-stop or enabling any output.

Relay and indicator behaviour

- GPIO23 HIGH drives the BC547 through 1 kohm. The transistor pulls IN1 LOW, so the relay board must be configured for low-level trigger.
- A 10 kohm base-to-emitter resistor keeps the transistor off while GPIO23 floats during boot.
- Relay de-energized: COM1 connects to NC1 and the yellow safe/inhibited indicator is on. Relay energized: COM1 connects to NO1 and the green permitted indicator is on.
- Install the 100 uF electrolytic across the fused 5 V rail before the E-stop branch split. Observe polarity. Do not put it after NC2 because stored energy could delay relay drop-out.
- Leave relay channel 2 unused in the first build.

7. Wire identification schedule

Label both ends with the W-number. Colour is a convenience, never proof of function. The schedule becomes the continuity checklist and the final as-built record.

ID	From	To	Signal	Colour	Control note
W01	Pi physical 1 (3V3)	ADS1115 VDD	3.3 V	Red	ADC supply
W02	Pi physical 6 (GND)	ADS1115 GND	0 V	Black	ADC return
W03	Pi physical 3 (GPIO2/SDA1)	ADS1115 SDA	I2C data	Green	No pull-up added initially
W04	Pi physical 5 (GPIO3/SCL1)	ADS1115 SCL	I2C clock	Yellow/blue	No pull-up added initially
W05	ADS1115 ADDR	ADS1115 GND	Address strap	Black	Selects address 0x48
W06	Pi physical 17 (3V3) via distribution node	Pressure pot measured high track end A	3.3 V	Red	Terminal must be measured
W07	Pi physical 14 (GND)	Pressure pot measured low track end C	0 V	Black/orange	Dedicated P1 ground; terminal measured
W08	Pressure pot measured wiper B	ADS1115 A0	0-3.3 V analogue	Yellow/green	Pressure emulator
W09	Same physical-17 3V3 distribution node	Flow pot measured high track end A	3.3 V	Red	Characterise this pot separately
W10	Pi physical 9 (GND)	Flow pot measured low track end C	0 V	Black/orange	Dedicated P2 ground; characterise this pot separately
W11	Flow pot measured wiper B	ADS1115 A1	0-3.3 V analogue	Blue/yellow	Flow emulator
W12	100 nF capacitor	ADS VDD to GND	Decoupling	N/A	Fit close to ADS module
W13	Pi physical 18 (GPIO24)	E-stop measured NC pair 1 terminal A	3.3 V logic	Yellow	NC fail-safe sense loop
W14	E-stop measured NC pair 1 terminal B	Pi physical 20 (GND)	0 V	Black	Normal closed = LOW
W15	Pi physical 16 (GPIO23)	1 kohm resistor	3.3 V command	Green	Resistor then transistor base
W16	1 kohm resistor	BC547 base	Base drive	Green	Verify E-B-C pinout
W17	BC547 base	10 kohm resistor to emitter	Pull-down	N/A	Keeps command off at boot
W18	BC547 emitter	Common ground point	0 V	Black	Logic/external supply reference
W19	BC547 collector	Relay IN1	Low-side input	Green	Relay must be set for LOW trigger
W20	External regulated +5 V	1 A fuse holder input	+5 V	Red	Separate supply, not Pi 5 V pin
W21	Fuse holder output	Fused +5 V rail	+5 V fused	Red	Install 1 A fast fuse
W22	Fused +5 V rail	E-stop measured NC pair 2 terminal A	+5 V	Red	Hardwired output inhibit
W23	E-stop measured NC pair 2 terminal B	Relay module DC+ or VCC terminal as marked	+5 V switched	Red	Relay loses power on E-stop
W24	External 5 V ground	Relay module GND	0 V	Black	External supply return
W25	External 5 V ground	Pi ground at one point	0 V reference	Black	Required for transistor interface
W26	Fused +5 V rail	Relay contact COM1	+5 V	Red	Indicator source
W27	Relay contact NC1	Yellow indicator +	+5 V when safe/off	Yellow	Default inhibited indication
W28	Relay contact NO1	Green indicator +	+5 V when permitted	Green	Energized indication
W29	Both indicator negative leads	External 5 V ground	0 V	Black	Verify polarity first
W30	100 uF capacitor + / -	Fused +5 V rail / ground	Bulk decoupling	N/A	Before E-stop split; observe polarity

8. Terminal verification register

Item	Do not assume	Multimeter or visual test	Record before wiring
E-stop	Printed NC claim or terminal position	Continuity in released and latched states for both independent pairs	NC1-A/B and NC2-A/B labels; released/pressed resistance
TE / ALPS pots	Centre terminal is always the wiper	End-to-end resistance and wiper-to-end sweep	A, B, C labels and rotation direction
BC547	Flat-face lead order across vendors	Part marking, datasheet package, diode-mode junction check	E, B, C on the actual pieces
Relay module	VCC/GND/IN order, trigger jumper, COM/NC/NO order	Read silkscreen; unpowered contact continuity; powered coil test only after supply check	Connector map and low-trigger setting
Indicators	Wire colour means polarity; internal resistor value	Current-limited 3-5 V polarity test	+/- wire and observed current/brightness
Fuse path	Holder clips make reliable contact	Continuity through installed 1 A fuse	Resistance and fuse identity
Breadboard	Power rails are continuous end-to-end	Continuity map, including any centre rail splits	Rail split drawing
External supply	Label voltage equals measured voltage	Measure unloaded and with relay load	Voltage, polarity, current rating

Before wiring each remaining stage

Use the available multimeter to identify terminals and record the result before connecting them to the Pi. Do not assemble or energize the E-stop, transistor, relay, indicator, or fused external-supply circuit until its specific acceptance gate has passed.

9. Step-by-step assembly and commissioning SOP

Commissioning sequence

Advance only after the evidence gate for the current stage passes

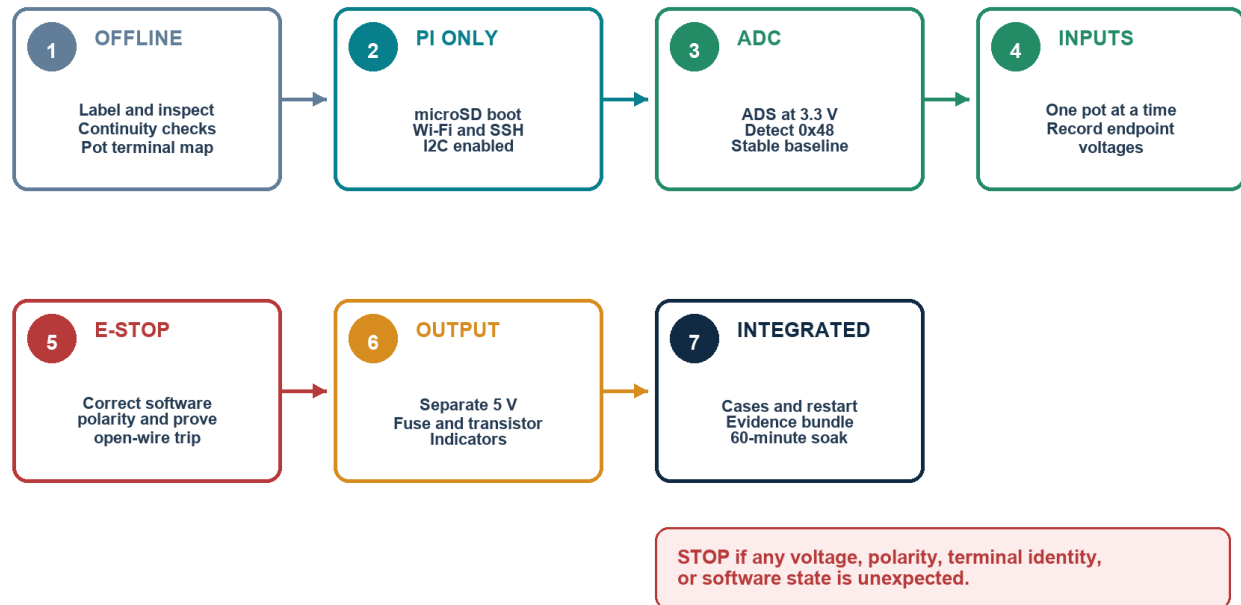


Figure 5. Controlled commissioning sequence. Each stage creates evidence before the next stage is allowed.

Gate	Action and acceptance
Stage 0 Work area	Both supplies disconnected; remove jewellery; dry non-conductive surface; label W01-W30; photograph the empty board.
Stage 1 Component acceptance	Map breadboard rails, pot terminals, E-stop pairs, relay terminals, transistor leads, indicator polarity, fuse continuity, and external supply voltage.
Stage 2 Pi-only boot	Flash Raspberry Pi OS, boot from PWR IN, join Wi-Fi, enable SSH and I2C, update hostname/time, and record OS plus software commit.
Stage 3 ADS-only	Power down. Wire W01-W05 and W12. Inspect. Power the Pi. Confirm I2C address 0x48 and stable open-channel behaviour. Power down.
Stage 4 One analogue channel	Wire pressure pot W06-W08. Check 0 V and 3.3 V rails before inserting ADS. Sweep slowly; record minimum/mid/maximum. Repeat for flow W09-W11.
Stage 5 Software safety	Implement NC-loop polarity and tests. Set relay_feedback_gpio to null initially. Create measured 0-3.3 V Wave 1 calibration. Prove output defaults safe after service start/restart.
Stage 6 E-stop input	Power down. Wire W13-W14 only. Validate released, pressed, and one-wire-disconnected states. All pressed/broken states must block the command.
Stage 7 Relay logic interface	With external supply still off, wire W15-W19 and W25. Confirm GPIO23 is LOW at boot and transistor is off. Verify no 5 V reaches the GPIO side.
Stage 8 Fused output circuit	Wire W20-W24 and W26-W30. Inspect polarity. Energize the external 5 V supply with Pi command forced safe. Yellow must be on, green off.
Stage 9 Integrated cases	Run normal, warning, trip, E-stop, broken loop, process restart, command rejection and 60-minute soak. Save logs, screenshots, wiring photos and measured results.

10. Software-to-wire contract

Software field	Current value	Physical meaning	Required action before bench
i2c_bus	1	GPIO2/SDA1 and GPIO3/SCL1	Keep
ads1115_address	72 / 0x48	ADDR tied to GND	Keep; confirm with I2C scan
valve_output_gpio	23	Pi physical pin 16 to BC547 interface	Keep; Wave 1 load is indicators only
emergency_stop_gpio	24	Pi physical pin 18 NC loop to ground	Keep; validated in automated polarity tests
emergency_stop_active_high	true	LOW = healthy; HIGH/open = active	Keep for the measured fail-safe NC loop
relay_feedback_gpio	null	Pi physical pin 22 remains disconnected	Keep disabled for the initial Wave 1 bench
output_active_high	true	GPIO23 HIGH commands transistor/low-trigger relay	Keep only after measured truth-table test
pressure raw range	Wave 1: 0.0-3.3 V	A0 pot is limited to the 3.3 V rail	Use rig.wave1.json; preserve raw voltage evidence
flow raw range	Wave 1: 0.0-3.3 V	A1 pot is limited to the 3.3 V rail	Use rig.wave1.json; preserve raw voltage evidence

Why GPIO25 stays disconnected

A relay contact may carry the external 5 V rail. Directly connecting that voltage to GPIO25 would damage the Pi. Feedback can be added later only as a 3.3 V dry-contact circuit or through an appropriately isolated/translated interface.

Current repository evidence

The adapter initializes its output safe and reads ADS1115 A0/A1. The Wave 1 configuration uses GPIO23 for output command, GPIO24 for the fail-safe NC E-stop loop, no GPIO25 feedback, and 0-3.3 V analogue scaling. Automated tests verify that LOW means healthy and HIGH/open means E-stop active before physical input commissioning.

11. Acceptance tests and evidence

ID	Test	Pass condition	Evidence
T01	Pi boot	Pi powered only through PWR IN; stable power; SSH reachable	Boot log + photo
T02	I2C	ADS1115 appears at 0x48 repeatedly	Command output + photo
T03	Pressure input	A0 sweep monotonic; min/mid/max recorded; no value above VDD	CSV + screenshot
T04	Flow input	A1 sweep monotonic; min/mid/max recorded; no value above VDD	CSV + screenshot
T05	Normal state	E-stop released; yellow on until a valid permit; no unsolicited relay action	Video + event log
T06	E-stop	Pressing E-stop removes relay power and software reports active	Video + timestamps
T07	Broken sense wire	Opening either NC1 conductor creates the same inhibited software state	Video + event log
T08	Threshold trip	Pressure above 4.2 bar equivalent rejects/de-energizes permit	CSV + decision record
T09	Invalid input	Out-of-range/stale input prevents output	Test result + event log
T10	Restart	After service/Pi restart the relay is de-energized and yellow is on	Video + boot log
T11	Indicator truth	NC/off = yellow; NO/on = green; states match dashboard	Photo/video matrix
T12	Soak	60 min with no unexplained reset, overheating, or state divergence	CSV summary + photos

Minimum evidence pack

- One labelled overhead wiring photograph and close-ups of Pi header, ADS, E-stop, transistor, fuse, relay, and indicator terminals.
- Multimeter acceptance sheet, measured supply voltages, potentiometer endpoint readings, and E-stop contact truth table.
- Exact configuration file, software commit hash, automated test output, I2C scan, CSV logs, screenshots, and short demo clips.
- A limitations statement preserving the dry-bench versus hydraulic boundary.

12. Troubleshooting by symptom

Symptom	Most likely checks	Do not do
Pi red LED only / no network	Confirm bootable microSD, correct PWR IN cable/supply, OS image, Wi-Fi config, router client list	Do not expect USB-A host cable to make the Pi a normal USB peripheral
ADS not at 0x48	Power off; check W01-W05, common ground, I2C enabled, ADDR strap, solder joints	Do not move wires while powered
ADC value jumps	Check breadboard rail continuity, 100 nF decoupling, short signal leads, secure pot wiper	Do not add random capacitors before a baseline log
Pot direction reversed	Swap only its two end terminals after power-off; keep wiper on A0/A1	Do not alter software sign to hide a wiring ambiguity
Relay always on	Disconnect external supply; verify LOW-trigger selection, 10 kohm pull-down, transistor E-B-C, GPIO boot state	Do not connect load until the input truth table passes
Pi resets when relay switches	Stop output; verify separate supply, common tie, supply voltage, wiring, decoupling and shorts	Do not power relay from Pi 3.3 V
E-stop reads backwards	Stop commissioning; correct NC-loop software polarity and rerun unit tests	Do not rewire to non-fail-safe semantics just to match current code
Green/yellow reversed	Power off; verify COM/NC/NO and indicator polarity	Do not assume relay terminal order from appearance

13. Build checklist and as-built sign-off

ID	Control	Result / measured value	Evidence filename
C01	All parts labelled and photographed		
C02	Breadboard rails mapped		
C03	Each pot high end / wiper / low end recorded		
C04	E-stop NC1/NC2 truth table recorded		
C05	BC547 E/B/C verified		
C06	Relay VCC/GND/IN and COM/NC/NO verified		
C07	External 5 V voltage/polarity/load rating verified		
C08	1 A fuse path continuity verified		
C09	NC E-stop polarity software change tested		
C10	Wave 1 0-3.3 V config frozen		
C11	W01-W30 point-to-point inspection complete		
C12	T01-T12 evidence archived		

As-built declaration

The circuit was assembled to this revision, all deviations were recorded, and all required gates were passed before integrated testing.

Role	Name	Date	Signature / review note
Builder	Chintan Dedhia		
Advisor review	Prof. Raj Kumar		

Revision history

Revision	Date	Status	Change
0.4	10 Sep 2026	Dual input verified	Recorded 4,457 paired samples; A0 reached 0-3.302 V and A1 reached 0-3.300 V; updated software and E-stop gates
0.3	10 Sep 2026	P2 ground allocation	P1 low terminal uses physical 14 GND; P2 low terminal uses physical 9 GND; both share the Pi ground plane
0.2	09 Sep 2026	As-built analogue correction	Use physical 17 for pot 3.3 V distribution and measured terminal convention A = high, B = wiper, C = ground
0.1	07 Sep 2026	Pre-commissioning	Initial controlled wiring design based on current inventory and repository contract

14. References

1. Raspberry Pi documentation, GPIO and 40-pin header:
<https://www.raspberrypi.com/documentation/computers/raspberry-pi.html>
2. Raspberry Pi configuration documentation, enabling I2C:
<https://www.raspberrypi.com/documentation/computers/configuration.html>
3. Raspberry Pi 3 Model A+ product page: <https://www.raspberrypi.com/products/raspberry-pi-3-model-a-plus/>
4. Texas Instruments ADS1115 data sheet: <https://www.ti.com/lit/ds/symlink/ads1115.pdf>
5. onsemi BC546/BC547/BC548/BC549/BC550 data sheet: <https://www.onsemi.com/pdf/datasheet/bc550-d.pdf>
6. Local project source: `work/certarig-poc/certarig_edge/hardware/raspberry_pi.py`
7. Local project configuration: `work/certarig-poc/config/rig.example.json`
8. Certarig Phase 3 hardware inventory register, baseline 06 Sep 2026.

Document control note

Manufacturer data sheets and measured terminal identities override illustrative diagrams. Update this document to an as-built revision after component acceptance and before Phase 3 submission.